

SOTKAMO KUOLANIEMI, Finland

WMO: 027390

Lat: 64.112N

Lon: 28.336E

Elev: 533

StdP: 14.42

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-15.8	-26.2	1.3	-20.8	-20.7	1.8	-15.8	16.7	22.3	15.8	21.9	1.9	160	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	79.7	64.3	75.9	62.5	72.5	60.9	67.1	74.8	65.1	72.8	62.9	69.8	6.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	92.9	70.1	62.1	85.0	68.1	59.9	78.6	66.1	31.9	74.5	30.3	72.9	28.7	69.8	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	13.6	12.1	DB	-26.4	82.6	7.8	6.0	-32.0	86.9	-36.6	90.4	-41.0	93.8	-46.7	98.1	(4)
(5)			WB	-26.0	70.3	7.5	3.7	-31.4	72.9	-35.8	75.1	-40.0	77.1	-45.4	79.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	12.0	15.8	23.7	35.2	47.6	55.3	61.5	57.7	49.2	36.6	28.8	20.0		
	DBStd	18.90	14.38	14.30	9.82	7.17	8.72	7.09	6.33	5.77	5.74	6.44	8.02	12.58		
	HDD50	5648	1178	957	816	445	151	27	2	7	82	416	636	931		
	HDD65	10254	1643	1377	1281	894	542	302	146	235	473	879	1086	1396		
	CDD50	929	0	0	0	2	75	185	359	247	60	1	0	0		
	CDD65	60	0	0	0	0	2	10	38	10	0	0	0	0		
	CDH74	519	0	0	0	0	39	110	303	67	0	0	0	0		
	CDH80	77	0	0	0	0	1	13	56	7	0	0	0	0		
	WSAvg	6.2	5.9	6.3	6.8	6.2	6.5	6.5	5.4	5.7	6.0	6.5	6.6	6.4		
	PrecAvg	26.6	1.8	1.5	1.4	1.2	2.3	2.8	3.5	3.0	2.5	2.4	2.1	2.1		
	PrecMax	34.1	3.0	2.6	2.0	2.3	4.3	6.0	6.9	6.5	3.9	4.9	4.3	3.7		
(15) Precipitation	PrecMin	21.7	0.4	0.3	0.4	0.4	0.4	1.1	1.1	0.9	0.5	0.7	0.2	0.6		
	PrecStd	3.0	0.6	0.7	0.4	0.5	1.2	1.2	1.3	1.5	1.0	1.1	1.0	0.7		
	0.4%	DB	35.6	40.2	46.6	63.0	78.1	82.0	84.6	80.3	68.3	54.2	47.5	38.6		
	MCWB	34.7	36.9	38.1	48.4	57.1	64.0	67.5	66.4	59.5	51.1	45.9	37.9			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	33.4	37.2	41.2	54.2	74.1	77.4	81.2	75.1	63.3	50.8	43.7	36.1		
	MCWB	32.5	34.8	35.5	44.1	56.6	63.1	66.2	63.2	56.2	48.3	42.4	34.9			
	5%	DB	32.0	34.6	38.0	49.8	69.3	72.3	77.2	71.5	60.2	48.1	40.6	34.4		
	MCWB	31.6	33.3	33.7	41.4	54.8	60.2	64.4	61.5	54.9	46.3	40.0	33.5			
	10%	DB	30.4	32.7	35.5	46.0	62.9	68.5	73.5	68.2	57.8	45.7	38.0	32.9		
	MCWB	29.9	31.9	32.6	38.9	51.0	57.4	62.4	60.0	53.8	44.0	37.4	32.3			
	0.4%	WB	34.8	37.6	38.9	49.3	60.7	66.8	70.8	68.9	60.2	52.2	46.6	38.3		
	MCDB	35.4	39.7	43.9	58.5	73.3	75.0	78.9	76.9	66.3	53.3	48.1	38.5			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.8	35.3	36.3	45.2	58.1	64.6	68.3	65.2	57.9	49.0	42.3	35.1		
	MCDB	33.2	36.5	39.6	53.2	69.5	75.0	76.2	71.8	61.6	50.3	43.2	35.8			
	5%	WB	31.7	33.4	34.7	41.7	55.6	61.5	66.3	62.9	55.8	46.4	40.0	33.5		
	MCDB	32.2	34.4	37.1	47.9	66.7	70.0	73.8	68.7	58.9	47.7	40.7	34.1			
	10%	WB	29.7	32.0	33.2	39.7	52.6	58.8	64.4	61.0	54.5	44.2	37.5	32.4		
	MCDB	30.3	32.8	35.0	44.3	62.8	66.2	71.7	66.7	57.3	45.7	37.9	32.8			
	MDBR	10.3	11.8	14.9	15.2	16.8	16.0	15.9	14.6	11.5	7.9	6.6	8.9			
	5% DB	MCDBR	6.8	8.3	13.6	21.8	26.8	22.1	21.8	19.9	15.0	10.5	7.9	6.4		
(36) Mean Daily Temperature Range	MCWBR	6.8	7.6	10.1	12.7	12.8	10.1	9.4	9.3	9.2	8.8	7.8	6.3			
	5% WB	MCDBR	6.9	8.6	10.8	18.7	24.3	20.1	19.6	17.0	13.2	10.0	7.8	6.4		
	MCWBR	7.0	8.1	8.7	11.4	12.5	9.6	9.6	9.2	8.5	9.3	7.7	6.4			
	taub	0.213	0.276	0.305	0.342	0.344	0.344	0.368	0.348	0.325	0.303	0.363	0.204			
(40) Clear-Sky Solar Irradiance	taud	1.962	2.107	2.157	2.266	2.453	2.478	2.427	2.468	2.521	2.386	2.662	1.963			
	Ebn at Noon	110	197	245	263	275	278	266	261	245	194	104	43			
	Edh at Noon	9	20	29	33	30	30	31	27	21	16	8	3			
	RadAvg	48	216	672	1170	1480	1652	1555	1169	642	235	55	18			
(44) All-Sky Solar Radiation	RadStd	6	46	69	126	162	110	162	120	69	38	11	2			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (31 neighbors)	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page